1.

$\vec{F} = q\vec{v} \times \vec{B}$

a) $\vec{V} = +Y$
$\vec{F} = -X$
$\therefore \vec{B} =$ into the Page

b) $\vec{V} =$ UP
$\vec{F} =$ into the Page
$\therefore \vec{B} = +X$

c) $\vec{V} = +X$
$\vec{F} = +Y$
$\vec{B} =$ Out of the Page

2.
$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

a) If closed:
Magnetic field increases into the Page inside Solenoid
Coil 1 lies to the left of the solenoid. The flux through it is increasing into the Page.
To oppose this, the induced current will create a field out of the Page → CCW current Counterclockwise

If Closed for a long time:
current becomes steady, no change in flux, no induced current

If opened:
Current stops, magnetic field decreases
Flux through coil decreases, field into the Page is weakening
Coil tries to maintain that field, induced current field into the Page CW current Clockwise

b) If closed:
Field increases out of the Page, induce field into the Page, CW current

If held closed:
no change, no induced current

If opened:
Flux decreases, induce field into the Page, CW current

c) If closed:
Field increases out of the Page, induce field into the Page, CW current

If held closed:
No change, no induced current

If opened:
Field decreases, induce field out of page, CCW current

3. Given:
$d = 2.20\ cm$
$\Delta t = 0.250\ s$
$B = 2.00 T$
$R = 0.0100\ \Omega$

a) $Area = \pi r^2 = \pi\left(\frac{d}{2}\right)^2 = \pi\left(\frac{0.0220}{2}\right)^2 = \boxed{3.80 \times 10^{-4}\ m^2}$

$emf = \frac{\Delta \Phi_B}{\Delta t} = \frac{BA}{\Delta t} = \frac{(2)(3.8 \times 10^{-4})}{0.250} = 3.04 \times 10^{-3}\ V = \boxed{3.04 mV}$

b)
$I = \frac{emf}{R} = \frac{3.04 \times 10^{-3}}{0.0100} = \boxed{0.304 A}$

c)
$P = I^2 R = (0.304)^2 \cdot 0.0100 = 9.25 \times 10^{-4} W = \boxed{0.925 mW}$

4. Given:
$\omega = 1875\ rad/s$
$\mathcal{E}_0 = 18.0V$
$A = 0.0100\ m \times 0.0300\ m = 3.00 \times 10^{-4}\ m^2$
$B = 0.640 T$

a) $\mathcal{E}_0 = NBA\omega \Rightarrow N = \frac{\mathcal{E}_0}{BA\omega} = \frac{18}{0.640 \cdot 3.00 \times 10^{-4} \cdot 1875} \approx \boxed{27 \text{ turns}}$

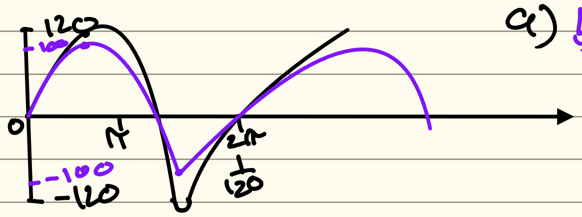
b) $T = \frac{2\pi}{\omega} = \frac{2\pi}{1875} \approx \boxed{0.00335\ s}$

5. Given:
$V_{primary} = 240\ V$
$V_{secondary} = 120 V$

a) $\frac{N_P}{N_S} = \frac{V_P}{V_S} = \frac{240}{120} = \boxed{2}$

b) $\frac{I_P}{I_S} = \frac{N_S}{N_P} = \boxed{\frac{1}{2}}$

6. Given:
$V(t) = V_0 \sin(\omega t)$
$V_0 = 120V, \omega = 240\pi\ rad/s$
Coil 3 has 1 turn, solenoid has 6 turns, they're aligned (same flux)



a) b)

c)

$V(t)=120\sin(240\pi t)=0 \Rightarrow \sin 240\pi t=0 \Rightarrow 240\pi t=n\pi \Rightarrow t=\frac{n}{240}$

So:

$t=0, \frac{1}{240}, \frac{2}{240}$

7. Given:

$I=0.100A$

$L=2.00mH=2.00\times10^{-3}H$

$\mathcal{E}=500V$

$$\mathcal{E}=L\frac{\Delta I}{\Delta t} \Rightarrow \Delta t=\frac{L\Delta I}{\mathcal{E}}=\frac{(2.00\times10^{-3})(0.100)}{500}=4.00\times10^{-7}s=400ns$$

8. Given:

$L=25H$

$I=100A$

$\Delta t=80.0ms$

a) $\mathcal{E}=L\frac{\Delta I}{\Delta t}=25\cdot\frac{100}{0.0800}=25\cdot1250=3125V$

b) $E=\frac{1}{2}LI^2=\frac{1}{2}25(100)^2=0.5\cdot25\cdot10{,}000=125{,}000J$

c) $P=\frac{E}{\Delta t}=\frac{125{,}000}{0.0800}=1.56\times10^6W=1.56MW$

d) using $L=\frac{\mu_0N^2A}{l}$, $\Rightarrow N^2=\frac{Ll}{\mu_0A}$

$\mu_0=4\pi\times10^{-7}$

$A=0.0100m^2$

$l=0.50$

$\therefore$

$$N^2=\frac{25\cdot0.50}{(4\pi\times10^{-7})\cdot0.0100}=\frac{12.5}{1.256\times10^{-8}}\approx9.95\times10^8 \Rightarrow N\approx31500 \text{ turns}$$

$A=0.01m^2$

$l=0.50m$

e) $X_L=\omega L=2\pi fL=2\pi\cdot10000\cdot25=1.57\times10^6\Omega$

9. Given:

$\tau=20ns=2.00\times10^{-8}s$

$R=5.00M\Omega=5.00\times10^6\Omega$

a) $\tau=\frac{L}{R} \Rightarrow L=\tau R=(2.00\times10^{-8})(5.00\times10^6)=0.100H$

b) $R=\frac{L}{\tau}=\frac{0.100}{1.00\times10^{-9}}=1.00\times10^8\Omega=100M\Omega$

c) $t=3.0ns$

In decay:

$I(t)=I_0e^{-t/\tau}=I_0e^{-3/20}\approx I_0e^{-0.15}\approx I_0\cdot0.861$

86.1%

10. a)

$V_{in} - V_R - V_C = 0$

$V_R = iR$

$V_C = V_{out}$

$i = C\frac{dV_{out}}{dt}$

So:

$V_{in} = iR + V_{out} = RC\frac{dV_{out}}{dt} + V_{out}$

b) $V_C + V_R - V_{in} = 0 \rightarrow V_{out} + RC\frac{dV_{out}}{dt} - V_{in} = 0$

c) $Z_{total} = R + \frac{1}{j\omega C}$

$\frac{V_{out}}{V_{in}} = \frac{Z_C}{R+Z_C} = \frac{\frac{1}{j\omega C}}{R+\frac{1}{j\omega C}} = \frac{1}{1+j\omega RC}$

d) Let $\omega = 2\pi f$

Let $f_c = \frac{1}{2\pi RC}$

$f = 100 f_c$

$\left|\frac{V_{out}}{V_{in}}\right| = \frac{1}{\sqrt{1+(100)^2}} = \frac{1}{\sqrt{10001}} \approx 0.0099995$

$f = 0.1 f_c$

$\left|\frac{V_{out}}{V_{in}}\right| = \frac{1}{\sqrt{1+(0.1)^2}} = \frac{1}{\sqrt{1.01}} \approx 0.995$

11. Given:

$R = 0.1\,k\Omega = 100\,\Omega$

$C = 1\,\mu F = 10^{-6}\,F$

$L = 10\,mH = 10^{-2}\,H$

a) $Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$

$\omega = 2\pi f$

$f = 0.1 f_0 \approx 159.15\,Hz$

b)

$\omega = 2\pi \cdot 159.15 \approx 1000\,rad/s$

$Z = \sqrt{100^2 + \left(0.01 - \frac{1}{1000 \cdot 10^{-6}}\right)^2} \approx 994.99\,\Omega$

$f = 10 f_0 \approx 15915\,Hz$, $\omega \approx 100\,000\,rad/s$

$Z \approx 100.4\,\Omega$

12. $V_{rms} = 120V$

a) $I_{rms} = \frac{V_{rms}}{Z}$

from above:

at $0.1 f_0$:

$$I_{rms} = \frac{120}{994.99} \approx 0.1205A$$

at $10 f_0$:

$$I_{rms} = \frac{120}{100.4} \approx 1.195A$$

b) Power

$P_{rms} = V_{rms} \cdot I_{rms} \cdot \cos\phi$

$\cos\phi = \frac{R}{Z}$, $= \frac{100}{994.99} \approx 0.1005$

$0.1 f_0$:

$P_{rms} = 120 \cdot 0.1205 \cdot 0.1005 \approx 1.45W$

$10 f_0$:

$P_{rms} = 120 \cdot 1.195 \cdot 0.996 \approx 142.8W$

13. $f_c = 1.4 MHz$

$f_a = 10k Hz$

Lower sideband: $f_c - f_a = 1.390 MHz$

Carrier: $1.400 MHz$

Upper sideband: $f_c + f_a = 1.410 MHz$

1.390 MHz, 1.400 MHz, 1.410 MHz

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1. Given:

$\Delta Q = 200 nC$

time: $\Delta t = 2\mu s$

$r = 1cm = 0.01m$

a) $I_d = \frac{\Delta Q}{\Delta t} = \frac{200 \times 10^{-9}}{2 \times 10^{-6}} = 0.1A$

$B = \frac{\mu_0 I_d}{2\pi r} = \frac{(4\pi \times 10^{-7})\, 0.1}{2\pi\, 0.01} = \frac{4 \times 10^{-8}}{0.01} = 4 \times 10^{-6} T$

b) displacement current

$I_d = 0.1A$ due to changing electric field

2. a) $d = \frac{c\Delta t}{2} = \frac{(3 \times 10^8)(10 \times 10^{-6})}{2} = \frac{3000}{2} = 1500m$

b) $R = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2$ $R = \left(\frac{1 - 1.78}{2.78}\right)^2 \approx 0.0786$

$\approx 7.9\%$

3.

a) Power = 1 kW = 1000 W

$A = 10 cm^2 = 10 \times 10^{-4} m^2 = 0.001 m^2$

$\text{Intensity} = \frac{P}{A} = \frac{1000}{0.001} = 1\,000\,000 \frac{W}{m^2}$

$1 \frac{MW}{m^2}$ @ 1 meter

b) $I_2 = I_1 \left(\frac{r_1}{r_2}\right)^2 = 1000000 \left(\frac{1}{2}\right)^2 = 250\,000 \frac{W}{m^2}$

$250 k \frac{W}{m^2}$ @ 2 meters

4.

a) Show $\theta_3 = \theta_1$ if $n_1 = n_3$

$n_1 \rightarrow n_2$

$n_1 \sin\theta = n_2 \sin\theta_2$

$n_2 \rightarrow n_3$

$n_2 \sin\theta_2 = n_3 \sin\theta_3$

$\therefore n_1 \sin\theta_1 = n_3 \sin\theta_3 \Rightarrow$ if $n_1 = n_3 \Rightarrow \theta_1 = \theta_3$

b) $M = \frac{25 cm}{f} = \frac{25}{12} \approx 2.08$

5) Given:

$\sigma = 200 \times 10^{-24} cm^2$

$d = 1 cm$

$n \approx 3.3 \times 10^{22}$

$\frac{I}{I_0} = e^{-n\sigma d}$, $n\sigma d = (3.3 \times 10^{22})(200 \times 10^{-24})(1) = 6.6$

$\frac{I}{I_0} = e^{-6.6} \approx 0.00136$

Passes through: $\approx 0.136\%$

absorbed: $\approx 99.86\%$

6.

a) 1 rad = 0.01 J/kg

$\text{Dose (rads)} = \frac{J}{kg} / 0.01 = \frac{0.250}{60} / 0.01 = 0.00417 / 0.01 = 0.417 \text{ rad}$

b) $\text{Dose} = \frac{0.250}{2.0} / 0.01 = 0.125 / 0.01 = 12.5 \text{ rads}$

c) $\text{Dose (Sv)} = \text{Dose (Gy)} \cdot RBE = (0.125 \text{ Gy}) \times 1 = 0.125 \text{ Sv}$

d) Yes, a moderate health risk bc it is above the 50mSv occupational exposure